

Mohammed Mamdouh

☎ 917-702-4473 | ✉ mmamdouh4370@gmail.com | [in linkedin.com/in/mohmamdouh](https://www.linkedin.com/in/mohmamdouh) | github.com/mmamdouh4370

EDUCATION

Georgia Institute of Technology

Dec. 2028

Incoming M.S. in Computer Science

University of Central Florida

May 2026

B.S. of Science in Computer Science, Honors | Minor in Intelligent Robotic Systems

3.8 GPA

Relevant Coursework: *Machine Learning, Computer Vision, DSA II, Linear Algebra, Statistics II, Robotics*

WORK EXPERIENCE

AI Engineer

May 2025 – Present

Universal Creative

Orlando, FL

- Developed and deployed an internal **speech to text transcription backend** with **speaker diarization** for up to three overlapping voices, incorporating advanced **noise filtering** to achieve high quality live transcriptions
- Designed and deployed a **Retrieval Augmented Generation (RAG)** system using **custom embeddings and vector search** over hundreds of manuals, supporting semantic search for **100+ internal users**
- Collaborated on a **perception based system** for dark ride environments, creating a **human in the loop annotation tool** to refine baseline model accuracy
- Built an automated **text classification pipeline** categorizing structured reports into **7+ internal classes**, integrating results with SQL and Excel based workflows used by **30+ employees**

Contract AI Engineer

Mar. 2025 – Aug. 2025

Contract/Freelance NDA

San Francisco, CA

- Developed a multimodal username classification system (**90.7% accuracy, 93.6% precision, < 1s latency**) using **sentence transformers, phoneme encoders, and DTW similarity** serving **10000+ users** in production
- Engineered a tagging pipeline combining **semantic, phonetic, and structural embeddings** via **Flask** optimized for low-latency inference and concurrent batch updates
- Scraped and labeled** a cross platform dataset of usernames to build a high quality baseline name to tag dataset

Research Assistant

Feb. 2025 – Jan. 2026

UCF FData Lab

Orlando, FL

- Benchmarked foundational and temporal **GNNs (DySAT, Roland, EvolveGCN, GraphAny)** on dynamic node/link/graph prediction using **PyTorch-Geometric** and **NetworkX**
- Developed a method using **topological persistence** to adapt **learning rate schedulers** across temporal graph snapshots, achieving an average **14% accuracy increase** on SOTA temporal GNNs

PROJECTS

BirdsEye (1st at Waymo & Assurant at ShellHacks 2025) | *YOLOv8, OpenCV, React, Flask*

- Developed a navigation app, integrating live **traffic camera feeds** from FDOT with user **Google Maps routes** to deliver lane-level **speed, congestion, and obstacle analysis** information
- Engineered backend pipelines for live video ingestion using **ffmpeg** and **YOLOv8** object detection to track vehicles and compute average lane speeds across multiple cameras

ACT: Autonomous Campus Taxi | *ROS2, Nav2, OpenStreetMap, C++*

- Built a physical **fully autonomous, rideable campus taxi**, enabling live **student pickup and dropoff** using onboard perception and navigation
- Designed the **global navigation stack** using **A*** routing over **OpenStreetMap** data, generating polyline routes for autonomous pickup and dropoff across a live campus environment
- Implemented **sensor based local path updating**, fusing **camera, LiDAR, and ultrasonic** inputs into real-time costmaps to adapt global routes for dynamic obstacle avoidance and safe navigation

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL, Java, Go, C#, HTML/CSS

AI/ML: PyTorch, TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, Librosa, NetworkX, OpenCV, YOLOv8

Web/Backend: React, React Router, Vite, Tailwind, Flask, FastAPI, Hono, Node.js, Express.js

Data/Infra: PostgreSQL, SQLite, MongoDB, Docker, AWS, Git, Linux, Vitest, Pytest